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/*
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Roll no: 07
Subject: Object Oriented Programming
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Date :

Problem statement:

Create a class named Shape with a function that prints "This is a shape". Create another class named Polygon inheriting the Shape class with the same function that prints "Polygon is a shape". Create two other classes named Rectangle and Triangle having the same function which prints "Rectangle is a polygon" and "Triangle is a polygon" respectively. Again, make another class named Square having the same function which prints "Square is a rectangle". Now, try calling the function by the object of each of these classes.

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*/
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#include<iostream>          // Header file included
using namespace std;        // Using standard namespace

// ----- Base Class Declaration -----
class Shape {               // Class Shape is declared
public:
    void output() {         // Member function is created
        cout << "This is a shape" << endl;
    }
};

// ----- Derived Class Polygon -----
class Polygon : public Shape { // Polygon class declared (inherits from Shape publicly)
public:
    void output() {         // Member function is created (function overriding)
        cout << "Polygon is a shape" << endl;
    }
};

// ----- Derived Class Rectangle -----
class Rectangle : public Polygon { // Rectangle class declared (inherits from Polygon)
public:
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    void output() {                // Member function is created (overriding)
        cout << "Rectangle is a polygon" << endl;
    }
};

// ----- Derived Class Triangle -----
class Triangle : public Polygon {    // Triangle class declared (inherits from Polygon)
public:
    void output() {                // Member function is created (overriding)
        cout << "Triangle is a polygon" << endl;
    }
};

// ----- Derived Class Square -----
class Square : public Rectangle {    // Square class declared (inherits from Rectangle)
public:
    void output() {                // Member function is created (overriding)
        cout << "Square is a rectangle" << endl;
    }
};

// ----- Main Function -----
int main() {                        // Main function starts

    Shape s;                        // Object of class Shape is created
    Polygon p;                      // Object of class Polygon is created
    Rectangle r;                    // Object of class Rectangle is created
    Triangle t;                     // Object of class Triangle is created
    Square sq;                      // Object of class Square is created

    s.output();                     // Function output() is called using object s
    p.output();                     // Function output() is called using object p
    r.output();                     // Function output() is called using object r
    t.output();                     // Function output() is called using object t
    sq.output();                    // Function output() is called using object sq

    return 0;                       // Program ends
}

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```
PS C:\Users\amitk\Desktop\cpp> cd "c:\Users\amitk\Desktop\cpp\" ; if ($?) { g++ shapes.cpp -o shapes } ; if ($?) { .\shapes }
This is a shape
polygon is a shape
Rectangle is a polygon
Triangle is a polygon
square is a rectangle
PS C:\Users\amitk\Desktop\cpp> █
```

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Output :

This is a shape  
Polygon is a shape  
Rectangle is a polygon  
Triangle is a polygon  
Square is a rectangle